Homework Solutions

Mathematics

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题 PartA1

Find the derivative of $x^3 + 2x^2 - 5x + 1$.

解. 对多项式逐项求导：$(x^3)' = 3x^2$

$(2x^2)' = 4x$

$(-5x)' = -5$

$(1)' = 0$

合并得 $3x^2 + 4x - 5$

Answer:

$$\boxed{3x^2 + 4x - 5}$$

题 PartA2

Evaluate the integral $\int x^2 \sin(x) dx$.

解. 使用分部积分，令 $u = x^2$，$dv = \sin(x)dx$，则 $du = 2x\,dx$，$v = -\cos(x)$

$\int x^2 \sin(x)dx = -x^2\cos(x) + 2\int x\cos(x)dx$

对 $\int x\cos(x)dx$ 再次分部积分：$u = x$，$dv = \cos(x)dx$，得 $x\sin(x) - \int \sin(x)dx = x\sin(x) + \cos(x)$

代回得 $-x^2\cos(x) + 2x\sin(x) + 2\cos(x) + C$

Answer:

$$\boxed{-x^2\cos(x) + 2x\sin(x) + 2\cos(x) + C}$$

题 PartA3

Compute the limit $\lim_{x\to 0} \frac{\sin(x)}{x}$.

解. $\lim_{x\to 0}\frac{\sin(x)}{x}=1$

Answer:

$$\boxed{1}$$

题 PartA4

Find the derivative of $e^{x^2}\cdot\ln(x)$.

解. 使用乘积法则：$(uv)'=u'v+uv'$，其中 $u=e^{x^2}$，$v=\ln(x)$

$u'=e^{x^2}\cdot 2x$（链式法则）

$v'=\frac{1}{x}$

因此导数为 $2xe^{x^2}\ln(x)+\frac{e^{x^2}}{x}$

Answer:

$$\boxed{2xe^{x^2}\ln(x)+\frac{e^{x^2}}{x}}$$

题 PartA5

Evaluate $\int_0^1\frac{1}{1+x^2}dx$.

解. $\frac{1}{1+x^2}$ 的原函数为 $\arctan(x)$

由牛顿-莱布尼茨公式：$\int_0^1\frac{1}{1+x^2}dx=\arctan(1)-\arctan(0)$

$\arctan(1)=\frac{\pi}{4}$，$\arctan(0)=0$

故积分值为 $\frac{\pi}{4}$

Answer:

$$\boxed{\frac{\pi}{4}}$$

题 PartB6

Solve the equation $x^2-5x+6=0$.

解. 因式分解：$x^2-5x+6=(x-2)(x-3)$

令 $(x-2)(x-3)=0$

得 $x=2$ 或 $x=3$

Answer:

$$\boxed{x = 2 \text{ 或} x = 3}$$

题 PartB7

Solve $x^3 - 6x^2 + 11x - 6 = 0$.

解. 试根 $x = 1$：$1 - 6 + 11 - 6 = 0$，故 $(x - 1)$ 是因子
多项式除法：$x^3 - 6x^2 + 11x - 6 = (x - 1)(x^2 - 5x + 6)$
继续分解：$x^2 - 5x + 6 = (x - 2)(x - 3)$
因此 $x^3 - 6x^2 + 11x - 6 = (x - 1)(x - 2)(x - 3) = 0$
得 $x = 1$，$x = 2$ 或 $x = 3$
Answer:

$$\boxed{x = 1, x = 2 \text{ 或} x = 3}$$

题 PartC8

Compute the determinant of the matrix $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$.

解. 矩阵 $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$
按行列式定义展开（SymPy 符号计算，逐步约化）
$\det(A) = -2$
Answer:

$$\boxed{-2}$$

题 PartC9

Find the eigenvalues of $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.

解. 矩阵 $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$

特征方程 $\det(A - \lambda I) = 0$，展开求解：

$\lambda = 3,\ \lambda = 1$

特征值: $\lambda_1 = 3,\ \lambda_2 = 1$

Answer:

$$\boxed{\lambda_1 = 3,\ \lambda_2 = 1}$$

题 PartD10

Solve the ODE $y' + 2y = e^{-x}$.

解. 解析得到微分方程: $2y(x) + \frac{d}{dx}y(x) - e^{-x} = 0$

SymPy dsolve 求解（1 阶）

通解: $y = \left(C_1 e^{-x} + 1\right) e^{-x}$（含任意常数 C1）

Answer:

$$\boxed{y = \left(C_1 e^{-x} + 1\right) e^{-x}}$$

Auto-generated: 10/10 problems solved by SymPy.